Materials, stress & strain

National 5 · Mechanisms & structures · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Calculate the strain.

- $\varepsilon = \Delta L \div L = 1.2 \div 2000$
- $\varepsilon = 6.0 \times 10^{-4}$ (no units)

No marks for: Mixing metres and millimetres

3 Quick check

- a) It is corrosion resistant, so it won't degrade in sea water
- b) None
- c) 32 N/mm²

4 Key terms

- 1 – B
- 2 – C
- 3 – A

5 Choose a suitable material from aluminium, mild steel and carbon fibre, and justify your choice.

- Carbon fibre (or aluminium)
- Justification linked to property: high strength-to-weight ratio, does not corrode in seawater

No marks for: Mild steel: it corrodes and is heavy

6 Calculate the original length of the rod.

- $L = \Delta L \div \varepsilon = 0.6 \div 0.0012$
- $L = 500$ mm

7 Calculate the minimum diameter of the wire.

- $A = F \div \sigma = 8000 \div 200 = 40$ mm²
- $d = \sqrt{(4A \div \pi)} = \sqrt{50.9}$
- $d = 7.1$ mm (round up for safety)